

B.Sc. (Hons.) Semester-III Mid-Term Examination, 2025-26**STATISTICS****Paper No.: STAMV31**

Date: 06.11.2025

Time: 4 pm – 5 pm

Instruction: There are two parts. Part-A is compulsory and attempt any one question from Part-B.

Part-A (Compulsory)

- (a) Define Bernoulli distribution. Let $0, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1$ be a random sample from Bernoulli population with parameter p ($0 < p < 1$). Write the likelihood function of p . [2]
- (b) If $M_X(t) = e^{4t + \frac{1}{2}t^2}$, then what will be the probability distribution of random variable Y which is defined as $Y = X - 4$? [1]
- (c) A bag contains 5 items out of which 2 are non-defective. Two items are being drawn at random by without replace method. Find probability that (i) both are defective, (ii) only one is defective. [2]
- (d) Let $X_1, X_2, \dots, X_n$ be a random sample from Poisson population with parameter λ . Write joint probability distribution of $X_1, X_2, \dots, X_n$. [1.5]
- (e) Let a bus arrives at Lanka at every interval of 5 minutes. If you are waiting for the bus, what is probability that your waiting time lies between 2 and 4 minutes? [1.5]
- (f) Let X_1, X_2 be a random sample from Normal distribution with mean μ and variance σ^2 . Find the variance of $2X_1 + 4X_2$. [2]
- (g) Define Geometric distribution and find its mean. [3]
-

Part-B (Attempt any ONE question)

Q. 2 If $X_1, X_2, \dots, X_n$ be a random sample from $N(\mu, 10)$, find maximum likelihood estimator of μ . Show that sampling distribution of the estimator obtained is also $N\left(\mu, \frac{10}{\sqrt{n}}\right)$. [7]

Q. 3 Let $X_1, X_2, \dots, X_n$ be random sample from Bernoulli distribution with parameter p ($0 < p < 1$). If $T = \sum_{i=1}^n X_i$ is an estimator of the parameter p , then what will be the sampling distribution of T ? Also, find standard error of estimator T . [7]

Q. 4 Distinguish between Bernoulli and Binomial distributions with examples. Find first four raw moments of Bernoulli distribution. [7]

[Click here to visit our website](#)

[Click here to visit our contributor](#)